

EURSKEM AI · ENGINEERING DOCUMENTATION

VOLUME 6

Production Readiness Assessment

A measured, evidence-based verdict on whether the platform is ready for production, with a prioritized issue register and a staged remediation roadmap.

Platform: Agentic Workflow Platform

Version: 0.1.0 · Generated: 2026-08-22

Built by Ayush Khandelwal

Typed workflows compiled to LangGraph · zero-token preflight · strict evidence lifecycle

1. Verdict

READY FOR PRODUCTION AFTER SPECIFIED FIXES.

The platform's security architecture, durability design (leases, durable checkpoints, idempotency ledgers, single-use tokens), deployment pipeline (immutable releases, automatic rollback, backups), and observability foundation are demonstrably production-grade. What blocks a clean launch today is a small, well-scoped set of regressions and gaps concentrated in the most recent work: a red CI at HEAD, one SSRF gap in the External Action node, one unimplemented MCP tool handler, and a shared-subprocess schema regression. None requires architectural change.

Basis. The assessment combines a full static audit with live measurements taken in this environment: the backend suite (1,918 passing / 11 failing), the zero-token preflight gate, the frontend type/lint/unit checks, secret-in-git scans, and a dependency review.

2. Baseline results (measured)

Check	Result	Classification
Backend import & test collection	OK · 1,932 collected	—
Backend pytest	1,918 pass · 11 fail · 3 skip	Regressions from recent commits
Workflow preflight (<code>--warnings-as-errors</code>)	4 workflows FAIL	Nullable / required-null fields
Frontend unit tests (vitest)	332 pass · 9 fail	Builder YAML round-trip drift
Frontend ESLint (<code>--max-warnings=0</code>)	26 errors	Hooks / fast-refresh
Frontend type check	4 errors	New test file + unused import
Secrets in git	None (only <code>.env.example</code> tracked)	OK
Dependency freshness	Current (jose 3.5, fastapi 0.140, httpx 0.28)	OK
CI at HEAD	All jobs would fail	Deployment blocked

3. Feasibility scorecard

Area	Rating	Evidence
Correctness	YELLOW	11 failing tests incl. one live workflow regression (sp01→w03) and one unimplemented MCP tool; all fixable and understood.
Security	YELLOW	Excellent baseline controls; one High SSRF gap in External Action, one Medium session-revocation gap.
Architecture	GREEN	Single runtime boundary, durable HITL, worker-safe coordination; no rebuild needed.
Database	GREEN	Versioned lease-locked migrations, indexes, session scoping; one config-drift trap (DB_NAME).
Performance	GREEN	Finite timeouts everywhere, caching, no unbounded queries found; minor sync-insert note.
Reliability	GREEN	Leases, replay, fallback chains, ambiguous-write handling.
Testing	YELLOW	Deep suite (1,932) but red at HEAD; frontend tests not gated in CI.
Observability	YELLOW	Strong logging/metrics/audit; no alert rules yet.
Deployment	RED	Pipeline design is excellent but cannot run because CI is red at HEAD (transient).
Maintainability	GREEN	Handbook docs, 4 TODOs in 80k LOC, consistent conventions.

Green = acceptable for production · Yellow = improvement required but manageable · Red = production blocker.

4. High-severity findings

ID	Title	Category	Production impact
REL-1	CI red at HEAD	Deployment	11 backend failures, 4 preflight fails, 26 lint errors, 4 tsc errors; CD gated on CI so nothing can ship.
SEC-1	SSRF in ExternalActionService	Security	No URL scheme/host validation; template-resolved URLs can reach internal services and cloud metadata.
COR-1	Unimplemented MCP tool handler	Correctness	get_quotations_for_account is declared but has no handler; runtime error when called.
COR-2	Shared sp01 schema change breaks w03	Correctness	3 end-to-end tests red; a shipped workflow's subprocess fails validation.

5. Medium-severity findings

ID	Title	Detail
COR-3	Candidates API shape regression	TypeError / {}-vs-[] mismatch in discovered-candidates handling.
COR-4	Builder YAML round-trip drift	Derived Start inputs written back into saved YAML (9 frontend tests red).
COR-5	Stale tests reference deleted workflow	3 modules point at removed horizon_proposal_hitl_pdf.yaml.
DB-1	DB_NAME divergence trap	Hardcoded DB_NAME vs settings.mongo_db can split-brain if MONGO_DB changes.
SEC-2	JWT role in token, no revocation	10h tokens; a deactivated user keeps access until expiry.
OBS-1	No alert rules	Metrics exist but nothing pages anyone.
REL-2	No global frontend 401 handling	Expired cookie surfaces as generic errors.
DEV-2	vitest not gated in CI	How the Builder drift shipped.

6. Low-severity findings

ID	Title
DEP-1	argon2 imported but undeclared; passlib declared but unused

CLN-1	Committed junk: _mounttest.txt, orphan ui/index.css, uiw/, root npm artifacts
SEC-3	Rate limiter fails closed (503) when Redis is down (documented tradeoff)
REL-4	Synchronous cost-ledger insert on the event loop (low volume)
DEV-3	Unpinned redis-stack-server:latest in the dev compose

No Critical findings: no confirmed data-loss, breach, or corruption path exists in the current code.

7. Staged implementation roadmap

Stage	Focus	Content
Stage 0	Protect the baseline	Freeze current build/test status as the reference point; keep the repo runnable after every step.
Stage 1	Production blockers	Fix the 11 backend failures and 4 preflight fails (TASK-1), add SSRF protection to External Action (TASK-2), restore the frontend gates and gate vitest in CI (TASK-3).
Stage 2	Correctness & reliability	Remove the DB_NAME divergence trap, add frontend session-expiry handling, lock the candidates API contract.
Stage 3	Code cleanup	Dependency hygiene (argon2/passlib), delete committed junk, pin the dev redis image.
Stage 4	Architecture	None recommended now; the residual in-memory fast paths are documented and Redis-dependent, not a scaling blocker.
Stage 5	Performance	Optional: async cost-ledger writes if LLM volume grows.
Stage 6	Test hardening	SSRF regression tests; explicit cross-user authorization tests.
Stage 7	Operational readiness	Alert rules, restore drill, session-policy decision, backup heartbeat.
Stage 8	Production validation	Re-run this entire battery plus staging smoke, backup/restore and rollback drills.

8. Production launch checklist

Item
☒ Production build passes (Docker image, compose config)
☐ Static/type checks pass — blocked by TASK-3
☐ Lint passes — blocked by TASK-3
☐ Critical tests pass — blocked by TASK-1
☐ Core end-to-end workflows pass — blocked by TASK-1
☒ Authentication verified (Argon2, JWT claim validation)
☒ Authorization verified (RBAC + session scoping)

- ☒ Resource ownership verified
- ☒ Secrets removed from source code
- ☒ Input validation verified (strict Pydantic schemas)
- ☒ Database migrations tested (lease-locked, idempotent)
- ☒ Database indexes reviewed
- ☒ Backup procedure exists (checksummed)
- ☐ Restore procedure tested — TASK-13
- ☒ Rate limiting reviewed
- ☒ Timeout behavior reviewed (finite on all calls)
- ☒ Retry behavior reviewed (Retry-After + fallbacks)
- ☒ Idempotency reviewed (operation ledger, single-use tokens)
- ☐ Third-party failure behavior tested — TASK-2
- ☒ Logging configured (structlog JSON + request IDs)
- ☒ Sensitive-data logging reviewed
- ☐ Alerting configured — TASK-12
- ☒ Health/readiness checks configured
- ☒ HTTPS/TLS configured (Caddy ACME, HSTS, CSP)
- ☐ CI pipeline verified — red at HEAD
- ☒ Deployment process documented (immutable releases)
- ☒ Rollback procedure verified (automatic on readiness)
- ☐ No unresolved High launch blockers — open

Items marked ☐ are the remaining conditions of the verdict. Once Stage 1 and Stage 7 items are complete and the Stage-8 battery is green, the verdict upgrades to **READY FOR PRODUCTION**.